

Flatworms

PHYLUM Platyhelminthes

CLASSES 6

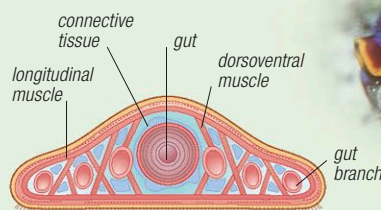
ORDERS 41

FAMILIES 424

SPECIES About 30,000

Flatworms are the simplest of all the animals that have bilateral symmetry. Their bodies are solid rather than containing an internal cavity, and they have no blood or circulatory systems, and no organs for exchanging gases with the

environment. Flatworms include many parasites, such as tapeworms and flukes, as well as a variety of free-living forms that are abundant in freshwater habitats and in the sea, particularly on rocky coasts and reefs.

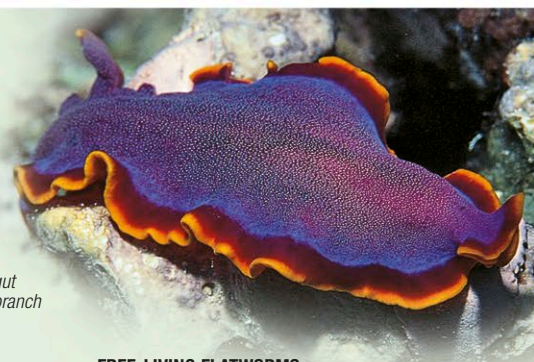


BODY SECTION

Flatworms have a simple body that is not divided into segments. The space between the internal organs is filled with spongy connective tissue.

FREE-LIVING FLATWORMS

Since flatworms rely on diffusion to take in oxygen, their bodies have to be thin. This requirement makes large species—such as this marine turbellarian—much more fragile than other animals of a comparable size.



Class Turbellaria

Turbellarians

Occurrence c.5,400 spp. worldwide; on rock and sediment surfaces in water, in damp soil, under logs

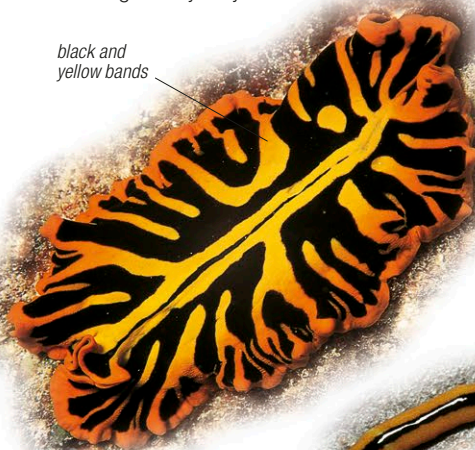


Turbellarian flatworms show a variety of different shapes. Although they have a distinct front end, the mouth is positioned some way back on the underside of the body. Most turbellarians are translucent, black, or gray, but some marine species—particularly ones that live in coral reefs—have brightly colored markings. They may be

less than $\frac{1}{32}$ in (1 mm) to 20 in (50 cm) long. The largest species are paper thin, and this helps them absorb oxygen directly from their surroundings. Unlike other flatworms, which are parasitic, most turbellarians are free living and have simple sense organs to help them negotiate their way around their habitat. Many of them move by creeping, but some of the bigger species swim by rippling their bodies. The majority of turbellarians are predators, feeding on other small invertebrates; some are parasitic, some are commensal, and a few depend on symbiotic photosynthetic algae. Almost all are hermaphrodites.

MARINE FLATWORMS

Some marine turbellarians are large and brightly colored. This species—*Psuedoceros dimidiatus*—is common on coral reefs in the Indian and Pacific oceans, and grows to 2 in (5 cm) long. It shows a variety of color schemes, including black and yellow traverse bands and a line of black blotches.



black and yellow bands

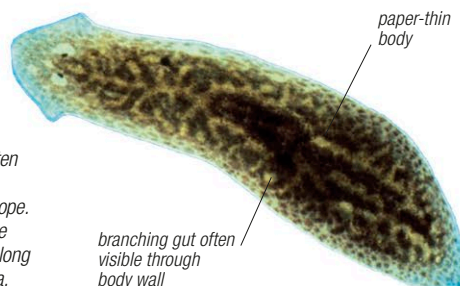
TERRESTRIAL FLATWORMS

Land-dwelling turbellarians often have elongated bodies, with head flaps bearing numerous small eyes. The *Bipalium* species, shown here, is one of the largest terrestrial forms, sometimes reaching 14 in (35 cm). Originally from the tropics, it has been introduced throughout the world via potted plants, and is now a widespread pest.

broad head with many eyes

BROWN FRESHWATER FLATWORM

Many turbellarians found in freshwater streams and ponds are drab in color, but their simple eyes—pigmented cups with light-sensitive retina but no lens—are often clearly visible. *Dugesia tigrina* is native to North America, but introduced into Europe. Like other turbellarians, they have a single opening to the gut and move by gliding along on a layer of microscopic hairs called cilia.



paper-thin body

branching gut often visible through body wall

Class Cestoda

Tapeworms

Occurrence c.4,000 spp. worldwide; within other animals, mostly cartilaginous fishes



Highly specialized for their parasitic way of life, tapeworms are usually less than $\frac{1}{32}$ in (1 mm) thick, but can be up to 100 ft (30 m) long. As adults, they live in the intestines of vertebrates—including humans—and they absorb food directly through their body wall. Tapeworms grow from a region just behind the head, forming a long line of segments called proglottids, each of which houses a complete reproductive system. As the proglottids age, they break away from the “tail” and leave the host in the feces, carrying eggs with them. If eaten by an appropriate host animal, the eggs develop—and the cycle begins again.



TAENIA TAPEWORMS

The widespread *Taenia* species—some over 33 ft (10 m) long—live in humans and other mammals. An adult specimen is shown above, while the close-up on the right shows the head, or scolex. It is equipped with hooks and suckers, with which the worm fastens itself to its host's intestines.



suckers on scolex

Class Trematoda

Flukes

Occurrence 18,000 spp. worldwide; within or on other animals



Like tapeworms, flukes are wholly parasitic, but they have a quite different shape. They are typically cylindrical or leaflike, with hooks or suckers at one or both ends, or on the underside. Their length ranges from less than $\frac{3}{8}$ in (1 cm) to, in exceptional cases, as much as 20 ft (6 m). Unlike tapeworms, flukes have mouths and digestive systems, and they often burrow through their hosts' tissues. While some attach themselves to internal organs, others are external parasites. Their life cycles are remarkably complex, involving a variety of larval stages, and up to 4 different hosts. The intermediate hosts are often mollusks, but the final host—which harbors the adult stage—is usually a vertebrate. Flukes can cause serious

diseases, including bilharzia, which affects millions of people in the tropics.

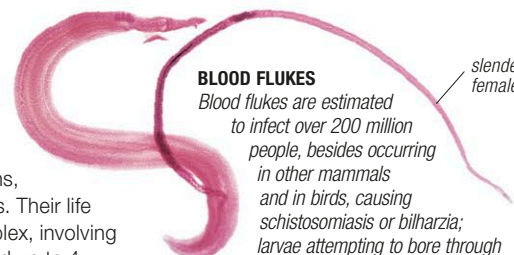
LIVER FLUKES

Several species of flukes live in the livers and bile ducts of mammals, including humans, causing severe damage and even death. The species shown above is the widespread sheep liver fluke, *Fasciola hepatica*, which grows to about $\frac{1}{4}$ in (3 cm) long. Its intermediate hosts are freshwater snails.



BLOOD FLUKES

Blood flukes are estimated to infect over 200 million people, besides occurring in other mammals and in birds, causing schistosomiasis or bilharzia; larvae attempting to bore through human skin also cause “swimmer's itch.” Unusually, the sexes are separate, although the male lives wrapped around the female. This tropical species, *Schistosoma mansoni*, is $\frac{3}{8}$ – $\frac{1}{2}$ in (1–1.5 cm) long.



slender female